MILITARY SPECIFICATION

GENERATORS, AMMONIA DISSOCIATED ATMOSPHERE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers atmosphere generators designed to produce an oxygen free, gas mixture composed of 75 percent hydrogen and 25 percent nitrogen for use in metallurgical processes.

1.2 Classification. The generators shall be of the following types and sizes as specified (see 6.2.1).

1.2.1 Types.

Type I - Gas Fired

Type II - Electrically Heated

1.2.2 Sizes. The size shall be based on cubic feet per hour (CFH) generator output of atmosphere, and should be selected from the sizes listed in Table I. If a size other than one listed in the table is required, the size in CFH shall be as specified.

TABLE I

Generator Output Capacity (Minimum)

Size	Cu Ft Per Hour	Size	Cu Ft Per Hour
1	75	6	1000
2	150	7	1500
3	300	8	2000
4	500	9	4000
5	750	10	6000

FSC 3424

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-3296	Forges, Furnaces, and Ovens (Exclusive of Space Heating and Cooking), Packaging of.

STANDARDS

Military

MIL-STD-130	Identification Marking of U S Military Property.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw Thread Standards for Federal Services.

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Circular 561 — Reference Table for Thermocouples.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

JOINT INDUSTRIAL COUNCIL (JIC)

EGP-1 — Electrical Standards for General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, Va. 22101.)

FACTORY MUTUAL ENGINEERING CORPORATION (FM)

Handbook of Industrial Loss Prevention, Chapter 37.

(Application for copies should be addressed to Factory Mutual Engineering Corporation, 1151 Boston-Providence Turnpike, Norwood, Mass. 02062.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1 and 6.3), the supplier shall furnish one complete generator of each type specified for first article inspection in accordance with Section 4.

3.2 Design. Generators provided in accordance with this specification shall be new and of the manufacturer's current design. The generators shall be designed to produce dissociated ammonia atmosphere by processing anhydrous ammonia through a catalytic retort at a temperature of at least 1750 degrees F. The generator shall consist of a vaporizer and heat interchanger system, refractory insulated heating chamber, catalytic retort, appropriate heating system for the type generator specified, and all necessary valves, piping, gages, flowmeters, safety devices, and control systems.

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3.2.1 Type I. Unless otherwise specified, the Type I generator shall be heated by natural gas having a heat content of 1000 BTU per cubic foot at a supply pressure of 6 - 8 inches water column. When required, the generator shall be designed and equipped to operate on either a butane or a propane fuel supply as specified (see 6.2.1).

3.2.2 Type II. The Type II generator shall be electrically heated and shall operate from a single source primary power supply. Electrical requirements shall be in accordance with 3.4.

3.2.3 Safety requirements. The generator shall be provided with safety features and devices for the protection of the generator and personnel. In addition to other safety requirements specified herein, the safety system of the generator shall conform to the requirements of FM "Handbook of Industrial Loss Prevention", Chapter 37.

3.2.4 Material. All material used in fabrication of the generator and related equipment shall be new and of the quality necessary to produce a generator to meet the requirements described herein.

3.2.5 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication of all sliding and rotating parts, except non-additive pack-seal bearings. Lubrication systems shall be designed and built to prevent contamination of atmosphere, components, material, or equipment. Lubrication points and systems shall be readily accessible, and identified in the operators manual with type of lubricant and instructions for application.

3.2.6 Accessibility. All parts subject to wear, breakage, or distortion, and all parts which require periodic maintenance, shall be accessible for adjustment and replacement, as applicable.

3.2.7 Maintainability. Item configuration and adjustment features will be designed to reduce maintenance operations

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to the simplest procedures compatible with a minimum of maintenance skills and common maintenance equipment. Instructions for disassembly, repair, adjustment, and reassembly will be concise and clearly defined.

3.2.8 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or equipment.

3.3 Construction. The generator shall be constructed so that when installed and connected to the required supply sources, it can be used for any purpose described herein. All parts and accessory equipment shall be functionally suitable for the purpose intended. Repaired parts shall not be used.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. In no event shall such processes as welding, peening, plugging, or filling with solders or pastes be used on castings or forgings to reclaim any defective parts for use on the generator.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where those operations are required in the original design. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners shall be installed with means for preventing loss of tightness. Such parts subject to removal or adjustment shall not be swagged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

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3.3.5 Painting. The generator shall be painted with heat resisting paint in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

3.4 Electrical equipment. The generator shall be designed and furnished complete with all electrical equipment, pre-wired where possible, for operation upon connection to source of power. All electrical equipment shall conform to JIC Electrical Standard EGP-1.

3.4.1 Power supply (Type II only). Unless otherwise specified (see 6.2.1), size 1 and 2 generators shall operate from a power supply of 240 volts, single phase, 60 Hz, and size 3 and larger generators shall operate from a power supply of 240 volts, 3 phase, 60 Hz.

3.4.2 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of a resin base.

3.4.3 Electromagnetic interference control. When specified (see 6.2.1), generators shall have electromagnetic interference control in accordance with the applicable requirements of MIL-STD-461.

3.5 Performance. The generator shall produce dissociated ammonia atmosphere having a chemical composition of approximately 75 percent hydrogen and 25 percent nitrogen. The undissociated ammonia in the produced atmosphere shall not exceed 0.05 percent. When specified (see 6.2.1), the generator shall be provided with a system to remove all traces of undissociated ammonia from the produced atmosphere. The dew point of the produced atmosphere shall be approximately minus 60 degrees F. The generator shall provide for selecting any desired atmosphere output flow rate from 25 percent to 100 percent of rated CFH and the atmosphere analysis shall remain constant at any desired flow rate.

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3.6 Components and systems.

3.6.1 Shell. The generator shell shall be fabricated of steel plate equal in thickness to at least 12 gauge. The base shall be fabricated of angle or channel iron. The entire unit shall be reinforced with steel members and shapes, and welded to form a structurally sound unit that may be skidded or moved without damage. If legs are used, they shall be provided with bearing plates.

3.6.2 Chamber lining. The combustion chamber lining shall consist of at least 2300 degrees F refractory backed up with insulating slab or preformed insulation. The quality and thickness of the lining shall be such that the outside chamber surface temperature will not exceed 140 degrees F when operating at 1750 degrees F. Powdered, shredded, or granular material shall not be used for insulating purposes.

3.6.3 Retort and catalyst. The retort shall be either fabricated inconel not less than 3/16 inch thick or cast alloy containing at least 35 percent nickel and 15 percent chromium. The catalyst shall be capable of withstanding a temperature of at least 2300 degrees F. The configuration of the catalyst shall prevent packing in the retort and shall provide sufficient surface area to assure complete dissociation of the ammonia.

3.6.4 Vaporizer and heat exchanger. The vaporizer system shall be sized and designed to assure an adequate supply of ammonia vapor to the dissociator for the size generator specified.

3.6.5 Heating systems.

3.6.5.1 Gas fired, Type I. The heating system for the Type I generator shall be complete and capable of maintaining a combustion chamber temperature of at least 1800 degrees F. The safety system of the gas fired generator shall conform to FM Handbook.

3.6.5.2 Electrically heated, Type II. Heating elements for the type II generator shall be of the ribbon or rod type

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with an alloy composition of 80 percent nickel and 20 percent chrome. Power density on the heating elements shall not exceed 12 watts per square inch of radiation surface at 1800 degrees F.

3.6.6 Instruments and controls. The temperature of the generator shall be automatically controlled by an electronic indicating controlling instrument equipped with thermocouple burnout protection. The instrument shall have an indicating calibrated scale length of not less than six inches and a temperature range of from 0 degrees F through at least 50 degrees F above the maximum operating temperature of the generator. The control system shall include a high pressure gauge to indicate the ammonia vapor supply pressure, a low pressure gauge to indicate pressure at the inlet to the retort; a flowmeter for indicating output flow of dissociated ammonia atmosphere, and all necessary pressure regulating, control and relief valves required for safe, efficient operation. Once the generator is started, operation shall be fully automatic.

3.6.6.1 Instrument panel. All instruments and controls shall be either integrally mounted or mounted in a separate floor mounted instrument panel. The instrument panel and mounting shall conform to JIC Standard EGP-1.

3.6.7 Thermocouples. Chromel-alumel type thermocouples shall be used as the temperature sensing device. The thermocouples shall conform to NBS Circular 561.

3.6.8 Excess temperature warning device. The generator shall be provided with a warning system to sound an alarm if the temperature varies more than 50 degrees F above or below operating temperature.

3.6.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.7 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate, marked in accordance

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with MIL-STD-130, shall be securely attached to each generator. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input characteristics and ratings
Gas BTU rating (Type I)
Output capacity (CFH)
Date of manufacture
Contract or order number
Federal Stock Number or Plant Equipment Code
U S

3.8 Identification plate. When specified (see 6.2.1), a separate identification plate of corrosion resistant metal permanently and legibly marked in letters 3/16 inches high with the caption "Departmental or Activity Identification" shall be securely attached to each generator. The plate shall have blank space for the Government to stamp 25 letters or numbers of the same size as the caption.

3.9 Workmanship. Workmanship of the generator and associated equipment shall be commensurable with the requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article when required in accordance with 3.1. Failure of the first article to pass the examination or one or more first article tests shall be cause for rejection. First article inspection shall consist of the examination in 4.5 and all tests in 4.6.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each generator prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.7. Each generator shall comply with all requirements.

4.5 Examination. The generator shall be examined for design, construction, materials, components, electrical equipment, safety conformance, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests. All tests shall be conducted at an ambient temperature of 77 degrees F $\pm$ 18 degrees F.

4.6.1 Circuit test. Each circuit of the electrical system shall be tested for continuity short circuits and faulty ground in accordance with JIC Standard EGP-1.

4.6.2 Electromagnetic interference control test. Equipment having electromagnetic interference control shall be tested in accordance with the applicable requirements of MIL-STD-461 to determine compliance with 3.4.3.

4.6.3 Operational and safety test. The generator shall be prepared for operation and operated in accordance with the

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manufacturer's handbook of instructions until the refractory and insulation has cured. With the generator operating at a temperature of at least 1750 degrees F, each safety control device and system on the generator shall be tested. Each system shall be tested by simulating a failure that normally would cause the device to operate. Each device or system shall perform its intended function.

4.6.4 Performance test. After the generator has been operated at 100 percent of rated capacity for at least three hours, an atmosphere composition analysis and a flow rate test shall be performed. The flow rate shall be at least the cubic feet per hour specified for the size generator under test. The atmosphere composition shall conform to the requirements in 3.5. All meters and instruments used in the test and analysis shall be of known accuracy.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing, and marking shall be in accordance with MIL-F-3296. Level of preservation, packaging, and packing shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Generators covered by this specification are primarily used to produce a dry, dissociated ammonia atmosphere for use in metal heat treating furnaces in brazing and metallurgical processes.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and the procurement documents shall specify the following:

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6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Type and size generator required (see 1.2).

c. First article, when required (see 3.1).

d. Type fuel, if different (Type I only) (see 3.2.1).

e. Electrical requirements, if different (Type II only) (see 3.4.1).

f. Electromagnetic interference control, when required (see 3.4.3).

g. Atmosphere scrubber, when required (see 3.5).

h. Repair parts, when required (see 3.6.9).

i. Nameplate, if different (see 3.7).

j. Identification plate, when required (see 3.8).

k. Level of preservation, packaging and packing required (see 5.1).

6.2.2 Contract data requirements. Technical data such as the manufacturer's standard operators manual, parts list, foundation and anchor bolt plans, and wiring diagrams will usually be required for delivery in connection with this specification. When technical data is required, it will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated into the contract.

6.3 Instructions for first article approval. When first article inspection and approval is required for equipment furnished under this specification, the solicitation

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should conform to the applicable requirements of Section I, Part 19 of the Armed Services Procurement Regulation.

Custodians:
Navy - SH
Air Force - 84

Review Activities:
Navy - AS, OS, SH, YD

User Activities:
Army - AV
Navy - MC
Air Force - 84

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